Warsaw University of Technology
Faculty of Mathematics
and Information Science

Master's diploma thesis

in the field of study Computer Science and Information Systems
and specialisation Artificial Intelligence

Multi-model LLM system with dynamic planning and model selection optimization

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WARSAW 2026

Abstract

ENGLISH TITLE

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Keywords: keyword1, keyword2, ...

Streszczenie

POLISH TITLE

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Słowa kluczowe: słowo klucz 1, słowo klucz 2, zażółć gęślą jaźń...

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1. Introduction

What is the thesis about? What is the content of it? What is the Author's contribution to it?

WARNING! In a diploma thesis which is a team project: Description of the work division in the team, including the scope of each co-author's contribution to the practical part (Team Programming Project) and the descriptive part of the diploma thesis.

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2. Example chapter

This TEX file is to be compiled with pdfLaTeX (it's just quick build in TeXMaker).

2.1. Example section

Definition 2.1 (Definition). A *definition* is a statement of the meaning of a term (a word, phrase, or other set of symbols).

2.1.1. Example subsection

It's the deepest deph of sectioning allowed by rector.

Definition 2.2 (Equation). In mathematics, an *equation* is a statement of an equality containing one or more variables.

Example 2.3. This is an example of an equation:

$$2 + 2 = 4. \tag{2.1}$$

Equation without a number:

$$2 + 2 = 4,$$

or:

$$2 + 2 = 4.$$

It is worthwhile to peruse other mathematical environments like *multline*, *align* and their versions with a star (, i.e. without numeration). The description of their use can be found at `https://texdoc.org/serve/amsldoc.pdf/0` starting from the end of the third page.

Equation (2.2) is false. References (and some other things) work properly after compliling TEX file twice.

$$\int_0^1 x\,dx = \frac{3}{2}. \tag{2.2}$$

Theorem 2.4 is a very interensting result.

Theorem 2.4 (Pythagoras' Theorem). Let c represent the length of the hypotenuse and a and b the lengths of the triangle's other two sides. Then:

$$a^2 + b^2 = c^2.$$

Proof. The proof has been presented in [1] and [2]. We can write then [1, 2]. □

Corollary 2.5. The use of the term *corollary*, rather than *proposition* or *theorem*, is intrinsically subjective.

Remark 2.6. You can find a rather comprehensive list of available symbols at `https://www3.nd.edu/~nmark/UsefulFacts/LaTeX_symbols.pdf`.

If you want to find a symbol by its shape, you can use the following site: `https://detexify.kirelabs.org/classify.html`.

Lemma 2.7 (Someone's Lemma). Ten lemat jest nie na temat.

Proof. Dowód przez indukcję. □

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2.2. Floats – tables and figures

Place labels after captions or you get the wrong labelling.

In Table 2.1 there are additional options for `table` and `figure` environments.

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Table 2.1: Additional options

symbol	effect
`h`	Place the float here, i.e., approximately at the same point it occurs in the source text (however, not exactly at the spot)
`t`	Position at the top of the page
`b`	Position at the bottom of the page
`p`	Put on a special page for floats only
`!`	Override internal parameters LaTeX uses for determining "good" float positions
`H`	Places the float at precisely the location in the LaTeX code. Requires the float package,[1] i.e., `\usepackage{float}`. This is somewhat equivalent to !ht.

Figure 2.1: Example figure – it has been drawn by LaTeX default tools

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3. The next chapter

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3.1. Matrices

Simple matrix:

$$\begin{matrix} a & b & c & d \\ d & e & f & g \\ 1 & 1 & 1 & 1 \end{matrix}$$

Matrix with parentheses:

$$A = \begin{pmatrix} a & b & c & d \\ d & e & f & g \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

Matrix with brackets:

$$\begin{bmatrix} a & b & c & d \\ d & e & f & g \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

You can also use more general environment:

$$\begin{matrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{matrix}$$

Matrix with braces:

$$\left\{\begin{matrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{matrix}\right\}$$

Definition 3.1. Let $A \neq \emptyset$, $n \in \mathbb{N}$. Every function $f\colon A^n \to A$ is called an *n-ary operation* or *działaniem* określonym na A. 0-ary operations are constant functions.

Definition 3.2 (Algebra). The ordered pair (A, F), where $A \neq \emptyset$ is a set and F is a family of operations defined on A, shall be called an *algebra* (or *F-algebra*). The set A is called *the set of elements*, *support* or *universe* of an algebra (A, F) and F is called *the set of elementary operations*.

Proposition 3.3. I state that, having passed to the limit, the only thing left me me is to camp at said limit or return, or, maybe, search for a pass or an exit to other areas.

Bibliography

[1] A. Author, *Title of a book*, Publisher, year, page–page.

[2] J. Bobkowski, S. Dobkowski, Title of an article, *Magazine X, No. 7*, year, PAGE–PAGE.

[3] C. Brink, Power structures, *Algebra Universalis 30(2)*, 1993, 177–216.

[4] F. Burris, H. P. Sankappanavar, *A Course of Universal Algebra*, Springer-Verlag, New York, 1981.

List of symbols and abbreviations

nzw.	nadzwyczajny
*	star operator
~	tilde

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List of Figures

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Spis tabel

2.1 Short caption 14

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List of appendices

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2. Appendix 2
3. In case of no appendices, delete this part.